

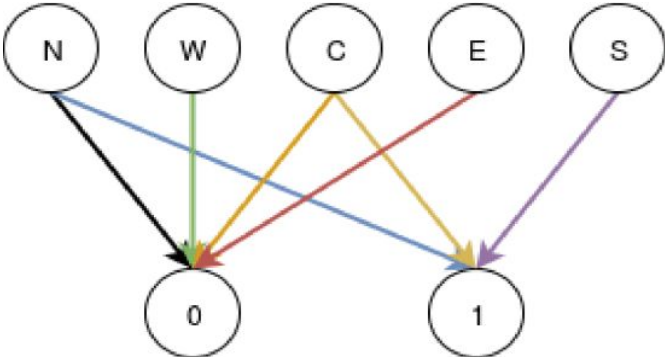
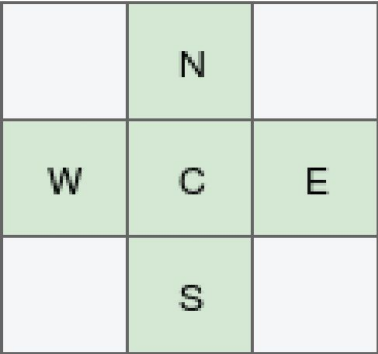
Local Computation

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Background

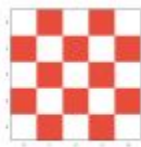
- Cellular Automata:**
 - A grid of cells, each with a state, that update simultaneously according to a local rule applied to their neighborhood
- Neural Cellular Automata:**
 - Traditional CA use a fixed hand-coded rule; Neural CA replaces that rule with a small neural network that runs identically in every cell



Type	Equation
Sigmoid	$f(x) = \frac{1}{1+e^{-x}}$
Hyperbolic tangent	$f(x) = \tanh(x)$
Sinusoid	$f(x) = \sin(x)$
Gaussian	$f(x) = ae^{-\frac{(x-b)^2}{2c^2}}$ a, b, and c are constants
Rectified linear unit	$f(x) = \max(0, x)$
Identity	$f(x) = x$
Clamped	$f(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$
Inverse	$f(x) = \frac{1}{x}$
Logarithmic	$f(x) = \log(x)$
Exponential	$f(x) = e^x$
Absolute value	$f(x) = x $
Hat	$f(x) = \begin{cases} 1 - x & x < 1 \\ 0 & \text{otherwise} \end{cases}$
Square	$f(x) = x^2$
Cube	$f(x) = x^3$

Background

- **CA-NEAT: Evolved Compositional Pattern Producing Networks for Cellular Automata Morphogenesis and Replication**
 - Evolving CPPN's to seed generate and replicate specific patterns



(a)



(b)



(c)

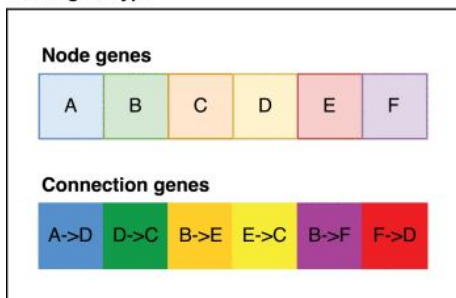


(d)



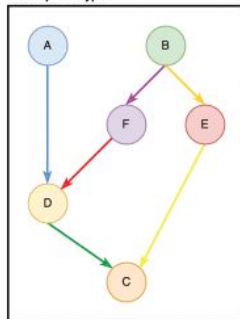
(e)

NEAT genotype

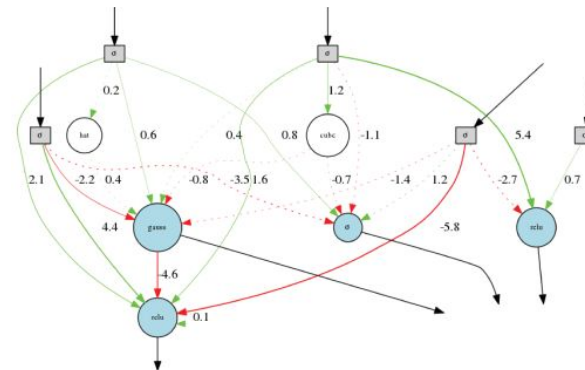


(a)

NEAT phenotype



(b)



(b)

Goals



Current Goal: Create a minimum working evolutionary pipeline evolving a rule set for a cellular automata problem (e.g. evolving a flag)

Project Goal: Investigate how self-repair emerges in evolved rule-sets (i.e. will the property emerge implicitly, which representation produces better repair, etc).

